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How to Improve Control Valve Performance with Positioners

7-9 minutes



As the final control element in most process control and automation systems, control valves are part of a complex system-within-a-system in which the valve positioner can be an integral component.

The control valve is made up of four principal sub-systems:

- The valve body, which includes the bonnet and piping inlet and outlet connections.
- The valve trim, plug, stem, cages, retainers and disc, used to control the flow through the valve body.
- The actuator, used to provide power to move the valve stem or shaft.
- The positioner, used to control the position of the valve stem and

provide linearization.

Do we need positioners on control valves?

To answer this question, we must answer some of the following questions: “Are diagnostics required on the control valve? Is criticality and variability an issue? Is the location of the valve in the field a long distance from the I/P or pneumatic controller?”

If the answer to any of those questions is yes, then a positioner is highly recommended. A positioner has several features that are performance related and can significantly improve the over-all control valve response to process set point and load changes.

One of the biggest problems of controlling the process is accurately positioning the valve stem (sliding stem) or shaft (rotary) during actuation.

The Actuator’s Function

The function of the actuator is to move the valve stem or shaft to some predetermined position that is called for by the controller (DCS or local controller). The actuators can control the valve stem position without a positioner. However, actuators are sometimes not very accurate and subject to many outside influences including dynamic stability (negative gradient), pulsation in the upstream flow from pumps and changing flow patterns due to piping geometry.

Actuators are initially linear, but over time the springs can weaken, and the effects of hysteresis and deadband will cause non-linear operation of the actuator. This could affect the dynamic error band of the actuator, decreasing its sensitivity, or resolution. The positioner corrects these non-linearities by being able to increase

air pressure to the diaphragm to get to the desired stem position through its feedback assembly.

Without position feedback, the actuator cannot accurately position the valve stem. This is worth repeating because the Cv of the valve is what controls the flow through the valve. Positioning the valve stem accurately will control the flow more accurately.

The Positioner's Function

As with most control valves today, the type and mode of positioners installed are reflected in the choice of actuation type, i.e., sliding stem or rotary actuators. Most of the pneumatic, electro-pneumatic and digital positioners are all capable of producing pneumatic outputs from 0 to 150 psig. This needs to be considered when choosing the type and style of positioner to use for a particular application.

Some of the attributes of the positioner are:

- Provides gain to the control signal by means of a higher proportional pressure (higher supply pressure) creating a higher force to the valve stem via the actuator, while its volume booster function provides the required capacity for the actuator volume resulting in a higher speed of response (pressurized and exhaust of actuator volume).
- Provides travel feedback of the valve stem for positioning accuracy.
- Provides linearization of the actuators stroke.
- Provides 0 psig loading pressure to the actuator when zero percentage signal is the input to the positioner.

The last function is of particular importance since the control valve

shut-off is related to the loading pressure of the actuator. This is true of an actuator that is reverse-acting, such as a spring-diaphragm, air to open, fail closed actuator. Since the spring is the closing force, the amount of force required to shut off the valve is equal to the lower bench-set multiplied by the actuator's effective area. The spring force is wound into the actuator to provide this force and this is commonly known as the lower bench set. Since any air pressure acting on the actuator will subtract away from the spring force, it is common practice to reduce this loading pressure to zero to increase the spring's shutoff force in a reverse-acting type actuator.

Getting More out of your Positioners

Digital positioners provide more than just accuracy. Besides providing the previously mentioned attributes of the pneumatic positioners, digital positioners can provide on-line diagnostics. This is extremely helpful with product data management and reliability issues when dealing with control valve-related process problems that show up from time to time in industrial applications.

The accuracy of digital positioners is excellent, with resolution up to 15,000 counts. With the new magnetic array type on some models, the measurement feedback is non-contact—no wear, not affected by vibration and can measure up to .004-inch accuracy.

Digital positioners also have adjustable gain, integral and rate adjustments. This makes them easier to deal with the different dynamics of the actuators involved; with some models, the setup wizard does well setting up these parameters, providing the correct actuator information is given during the setup.

Running diagnostics on the valve package before placing it into service is a very good idea. In this manner, you have a baseline diagnostic to refer to later if, and when, you have problems with the controllability of the valve package. In most cases, the information provided by the DVCs is both graphical and numerical. With some training, the user can find small problems—such as seat wear, worn or broken springs, leaking diaphragms and a host of others—before they get to be big problems.

Conclusion

The reasons to use a positioner on control valves in most applications include:

- Positioners create gain to the actuation system on pneumatic-spring diaphragm actuators by enabling higher supply pressures to be used.
- Linearizes actuator performance by decreasing deadband and hysteresis effects of the actuator.
- Provides positive stem position feedback for accurate stem positioning.
- Increases stem position resolution by utilizing higher supply pressure and higher spring rates because of the installation of the positioner.
- Provides 0 psig loading pressure to the actuator's 0% input signal. This helps the actuator, on air to open, fail close, to utilize the lower bench set wind-in to provide the necessary shut off force.

Digital positioners with diagnostics can help support efforts for performance diagnostic maintenance (PDM) that are critical to

control valve life cycle and overall maintenance of the control valves.

While all positioners, pneumatic, electro-pneumatic or digital are not equal, in my opinion any positioner is better than no positioner.

Any decision to use a positioner on a valve should be made with your control valve specialist or your local business partner who provides your control valves. They can advise you on the proper types of positioners and installation technique critical to optimizing the control valve.

This article is an excerpt from a comprehensive technical article by ***Louis Reeves****, retired electrical and instrumentation technician, control valve asset management at International Paper. Access the entire article HERE.*

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